

Fat apoptosis through targeted activation of caspase 8: a new mouse model of inducible and reversible lipoatrophy

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We describe the generation and characterization of the first inducible 'fatless' model system, the FAT-ATTAC mouse (fat apoptosis through targeted activation of caspase 8). This transgenic mouse develops identically to wild-type littermates. Apoptosis of adipocytes can be induced at any developmental stage by administration of a FK1012 analog leading to the dimerization of a membrane-bound, adipocyte-specific caspase 8–FKBP fusion protein. Within 2 weeks of dimerizer administration, FAT-ATTAC mice show near-knockout levels of circulating adipokines and markedly reduced levels of adipose tissue. FAT-ATTAC mice are glucose intolerant, have diminished basal and endotoxin-stimulated systemic inflammation, are less responsive to glucose-stimulated insulin secretion and show increased food intake independent of the effects of leptin. Most importantly, we show that functional adipocytes can be recovered upon cessation of treatment, allowing the study of adipogenesis *in vivo*, as well as a detailed examination of the importance of the adipocyte in the regulation of multiple physiological functions and pathological states.

Current murine models of lipodystrophy or lipoatrophy have contributed substantially to our understanding of adipocyte physiology^{1–4}, as well as to our understanding of diseases associated with excessive adipose mass, such as atherosclerosis, diabetes and hypertension. One consistent disadvantage to current lipoatrophic mouse models is the lack of inducibility. In addition, loss of adipose tissue in the current mouse models is irreversible such that the constitutive loss of adipocytes leads to lipotoxicity^{5–8}, which makes the distinction of the acute loss of adipokine-induced effects versus the prolonged pathologic changes associated with lipoatrophy difficult. Here, we describe the generation and characterization of the FAT-ATTAC mouse, a transgenic mouse in which we can induce the ablation of both white and brown adipocytes within 1–2 weeks of treatment at any stage of development and without substantial secondary effects in other tissues.

RESULTS

Generation and characterization of an inducible fatless mouse

The p20 and p10 catalytic domains of human caspase 8 were fused to serial FKBPv (Phe36Val mutant FKBP) domains that can be tethered using regulated homodimerization with introduction of a FK1012 analog, AP20187. AP20187 binds with ~1,000-fold greater (subnanomolar) affinity to FKBPv than to endogenous FKBP. To effectively increase local concentrations of the fusion protein, a myristoylation site was included to provide membrane attachment for the caspase 8–FKBP (Fig. 1a). The adipocyte-specific *Fabp4* promoter was used to drive expression of fusion protein in stably transfected 3T3-L1 preadipocytes. Treatment with AP20187 results in widespread apoptosis with loss of adipocyte membrane integrity and lipid contents (Fig. 1b). Similar to previous studies with caspase 8 in cardiac myocytes⁹ and caspase 3 in hepatocytes¹⁰, we generated transgenic mice that carry this transgene. Upon northern analysis, only one line of six that had germline transmission showed marked expression at the mRNA level (data not shown) and this line was used for all subsequent experiments. Transgene expression was limited to all white adipose depots examined, as well as interscapular brown adipose tissue, but was completely absent in all other tissues (Fig. 1c), which was further verified by RT-PCR (data not shown). Furthermore, western blot analysis using FLAG-specific antibodies indicated that fusion protein expression was only detectable in white and brown adipose tissue (data not shown). Based on the strategy used, we refer to this mouse as the FAT-ATTAC mouse.

To determine whether treatment with AP20187 dimerizer was equally effective in promoting adipocyte apoptosis *in vivo*, we analyzed inguinal and epididymal adipose depots from treated FAT-ATTAC mice. Short-term treatment with dimerizer (7 d) is sufficient to cause disruption of local adipose tissue structures with associated infiltration of macrophages and fibrosis (Fig. 2a); the number of cells expressing the macrophage-specific F4/80 antigen is markedly increased in adipose from FAT-ATTAC mice in close proximity to adipocytes undergoing apoptosis (Fig. 2a). Longer courses of dimerizer treatment (14–28 d) result in progressive disruption of adipose architecture with concomitant apop-

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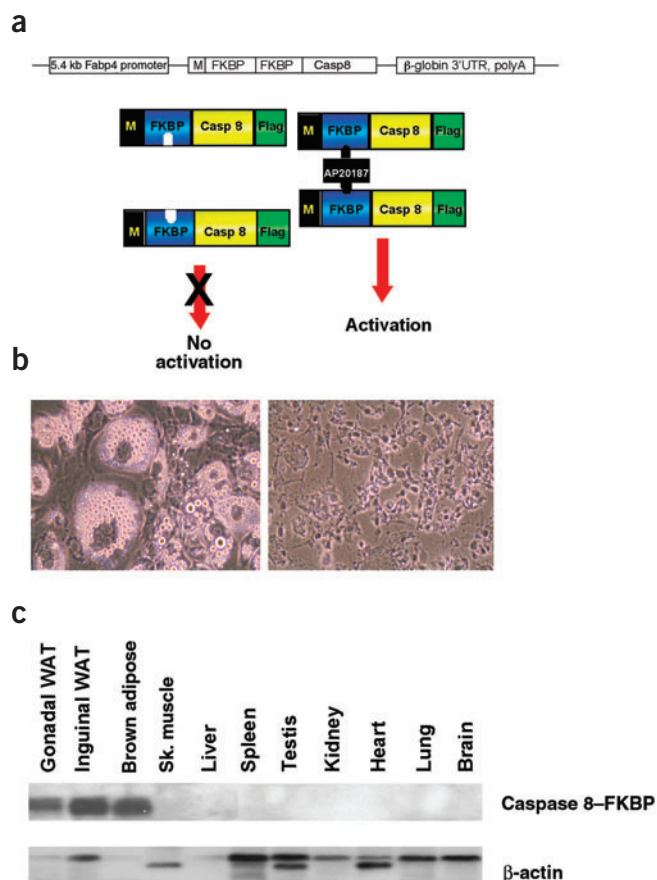


Figure 1 Adipocyte-specific expression of the caspase 8-FKBP transgene *in vitro* and *in vivo*. **(a)** Linearized expression construct introduced into blastocysts for transgenic expression of myristoylated caspase 8-FKBP fusion protein (top), and mechanism of activation through forced dimerization of adjacent FKBP molecules by dimerizer (bottom). **(b)** 3T3-L1 preadipocytes stably transfected with expression construct above were differentiated into adipocytes, then exposed to dimerizer (right) or vehicle (4% ethanol; left) treatment for 36 h. Note complete loss of lipid contents and adipocyte morphology in dimerizer-treated cells. **(c)** Tissue northern analysis on various tissues from 6-week-old male FAT-ATTAC mice. We hybridized total RNA with probes to human caspase 8 (top) or mouse β -actin (bottom) for normalization purposes. Samples were unequally loaded to show absolute absence of transgene in nonadipose tissues. Caspase 8-FKBP transgene expression is limited to both brown and white adipose tissue (WAT).

significant differences in fasting serum insulin, glucose, triglyceride or free fatty acid levels throughout the course of dimerizer treatment (data not shown). Upon cessation of treatment, adiponectin and resistin levels rebounded, indicating that adipocyte progenitor cells remain intact and the effects of the widespread adipocyte apoptosis are reversible. Progressive weight gain confirmed the global, robust return of adipose stores (**Fig. 2c**). Six weeks after removal from dimerizer treatment, FAT-ATTAC mice were heavier and showed increased body fat relative to pretreatment state as assessed by magnetic resonance imaging (MRI; **Fig. 2d**). Although metabolically identical to wild-type littermates before dimerizer treatment (**Fig. 2e**), fatless FAT-ATTAC mice were markedly glucose intolerant (**Fig. 2e**), consistent with decreased circulating adiponectin levels. Glucose intolerance persisted 6 weeks after cessation of dimerizer treatment (**Fig. 2e**), an observation that may be attributed to the fact that adiponectin levels only recover to approximately 50% of baseline levels. Circulating leptin levels are substantially decreased in fatless mice as well (**Fig. 2f**), leading to marked hyperphagia (~50% increased food intake).

FAT-ATTAC *ob/ob* mice

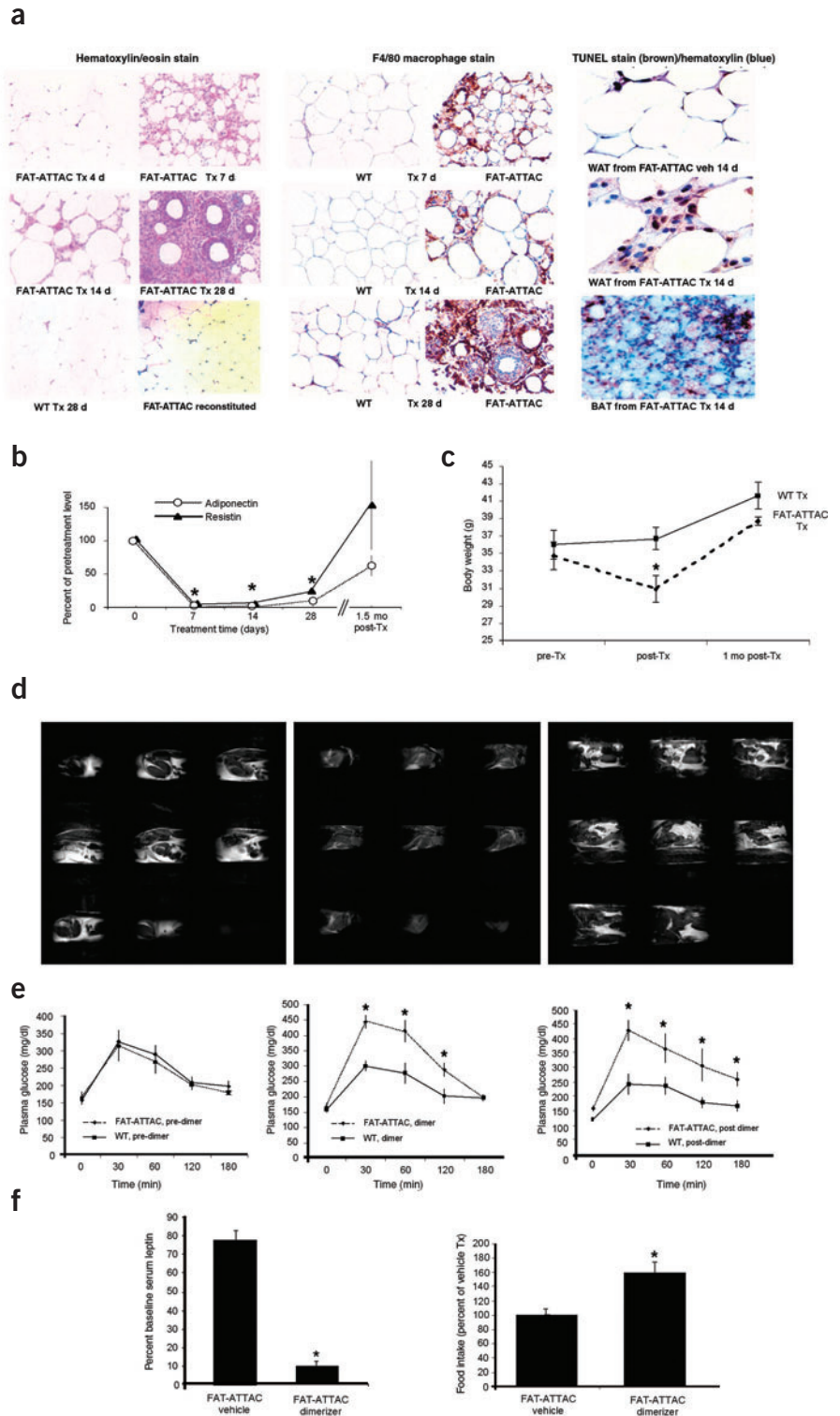
To determine whether the metabolic phenotype and associated changes in feeding behavior of the FAT-ATTAC mouse were attributable solely to the absence of leptin, an adipocyte-derived satiety signal, we generated FAT-ATTAC *ob/ob* mice and compared them to leptin-deficient *ob/ob* littermates. We observed similar weight curves for FAT-ATTAC *ob/ob* mice and *ob/ob* littermates throughout development (data not shown). After 10 weeks of dimerizer treatment, FAT-ATTAC *ob/ob* mice weighed nearly 30 g less than treated *ob/ob* littermates (**Fig. 3a**). Upon dissection, these FAT-ATTAC *ob/ob* mice showed near-complete absence of white adipose tissue (**Fig. 3b**). This finding was confirmed by MRI (data not shown). Histologically, there was an absence of lipid-laden adipocytes in white adipose depots (data not shown), similar to fatless FAT-ATTAC mice in the FVB background. Circulating levels of adipocyte-specific secretory proteins adiponectin and resistin were markedly reduced, whereas MCP-1 levels were elevated nearly fourfold (**Supplementary Fig. 3** online).

Fatless FAT-ATTAC *ob/ob* mice on dimerizer showed progressive increases in daily food intake (**Fig. 3c**), resulting in a 2.5-fold increase of caloric intake per gram body weight in the fatless *ob/ob* mice. There was no evidence of malabsorption of nutrients. Decreased body weight in the presence of excess caloric intake is indicative of elevated metabolic rate. To indirectly assess metabolic rate, we surgically implanted temperature-sensitive transmitters into the interscapular region of anesthetized mice and monitored core body temperature in conscious, unrestrained animals in either the fasted or fed state. Fatless FAT-ATTAC *ob/ob* mice showed a significantly higher core body temperature than

apoptosis of lipid-laden adipocytes. TUNEL-positive nuclei were present in both white and brown adipose tissue, whereas no substantial TUNEL staining could be observed in nonadipose tissues or in adipose tissue from control mice (**Fig. 2a**). TUNEL-positive cells were also positive for activated caspase 3, the downstream effector of the caspase cascade (data not shown). Unusual granuloma-like structures were abundant in FAT-ATTAC mice treated for 28 d. Though the exact nature of these structures is unclear, cells surrounding the central lumen express neither perilipin nor F4/80, suggesting that they are neither adipocytes nor macrophages and may represent reactive fibrosis to widespread adipocyte apoptosis (**Supplementary Fig. 1** online). Adipose tissue is morphologically normal in vehicle-treated transgenic mice and dimerizer-treated wild-type littermates and no detectable histological changes were present in any other tissue, confirming the adipocyte-specific expression of the transgene. Furthermore, there was no evidence of lipid accumulation in the liver (**Supplementary Fig. 2** online), suggesting that widespread adipocyte apoptosis does not result in the accumulation of triglyceride stores in peripheral tissues without persistently elevated serum triglyceride levels. Most importantly, 6 weeks after cessation of dimerizer treatment, adipose tissue from FAT-ATTAC mice appeared histologically identical to untreated mice, indicating that in the absence of continued dimerizer treatment, endogenous preadipocytes can successfully differentiate, enabling adipose tissue regeneration.

The absence of adipocytes observed histologically is mirrored by marked reductions (>95%) of the adipocyte-specific secretory proteins adiponectin and resistin by day 7 of treatment (**Fig. 2b**), and serum concentrations remain at near-knockout levels throughout the course of dimerizer treatment. Despite changes in circulating adipokines, fatless FAT-ATTAC mice in the genetic background of FVB mice showed no

Figure 2 Dimerizer treatment in FAT-ATTAC animals results in adipocyte apoptosis and reversible loss of adipocyte-specific markers *in vivo*. **(a)** Hematoxylin/eosin stain of adipose depots after initiation of dimerizer treatment (left panels). Immunohistochemistry for F4/80 indicates macrophage infiltration in adipose as early as 7 d after initiation of dimerizer treatment; the number of F4/80(+) cells increase with longer treatment periods (middle panels). Brown, F4/80; blue, hematoxylin counterstain. TUNEL stain (right panels) indicates that proportion of cells from both white (WAT) and brown (BAT) adipose tissue from dimerizer-treated FAT-ATTAC mice stain positively (brown) after 14 d of treatment; no TUNEL-positive cells were present in WAT or BAT from vehicle-treated or control animals (data not shown). **(b)** Serum levels of total adiponectin and resistin decrease to less than 5% baseline values within 7 d of dimerizer treatment. Within 1.5 months after cessation of dimerizer treatment, serum adiponectin and resistin levels rebounded to ~40% and 150% of baseline levels, respectively. **(c)** FAT-ATTAC mice (dashed line) decrease mean total body weight by 12% (~2 g) during treatment period of 21 d. Upon cessation of treatment for 6 weeks, FAT-ATTAC mice regained lost body weight and overshoot pretreatment values. **(d)** MRI analysis before treatment (left), after dimerizer treatment for 3 weeks (middle) and 6 weeks after cessation of dimerizer treatment (recovery phase, right) from representative FAT-ATTAC mouse, showing complete loss of fat signal (white) after dimerizer treatment but robust recovery of ablated adipose in the absence of dimerizer. **(e)** Oral glucose tolerance test in FAT-ATTAC and wild-type littermates before (left), during (center) and after treatment (right). Fatless mice show significantly ($P < 0.05$) impaired glucose tolerance that persists after regeneration of fat in FAT-ATTAC mice. **(f)** In FAT-ATTAC mice treated with either vehicle or dimerizer for 3 weeks, serum leptin levels drop (~10% of baseline levels) corresponding with an increase in food intake. WT, wild-type; Tx, treatment; mo, month.



ob/ob littermates (Fig. 3d), indicative of elevated metabolic rate.

Despite the massive weight loss and increased metabolic rate of these animals, complete loss of adipose function in the FAT-ATTAC *ob/ob* mice seemed to exacerbate the metabolic disorders inherent to this mouse model. Fatless FAT-ATTAC *ob/ob* mice showed further elevations over already high levels of fasting and fed plasma glucose (60% and 25% increase, respectively; Fig. 3e) and fasting and fed triglycerides (100% and 300% increase, respectively; Fig. 3f) relative to *ob/ob* littermates, whereas there was no significant difference in free fatty acids (Fig. 3g). In addition, histology of *ob/ob* versus fatless FAT-ATTAC *ob/ob* livers showed increased steatosis in the FAT-ATTAC *ob/ob* mouse (Fig. 3h). Furthermore, we showed that thiazolidinedione treatment is ineffective in ameliorating the diabetic phenotype of *ob/ob* mice in the absence of functional adipose tissue

(Supplementary Fig. 4 online). Together with the data generated from the fatless FAT-ATTAC mice in the FVB background, this suggests that the severity of the metabolic complications associated with fat loss is increased with the quantity of lipid that is displaced and that mere deletion of adipose tissue is not the simple solution to type 2 diabetes.

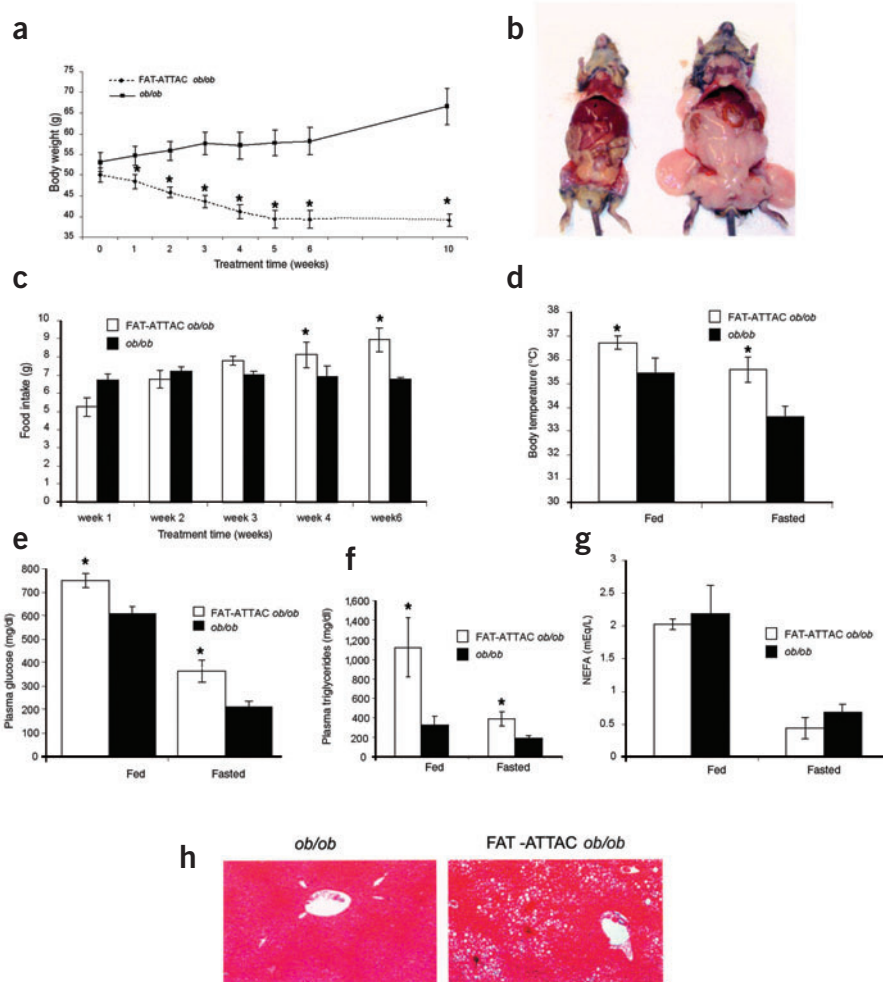


Figure 3 Effects of acute lipatrophy in the FAT-ATTAC *ob/ob* mouse model. **(a)** Decreased body weight with dimerizer treatment in FAT-ATTAC *ob/ob* mice as compared with *ob/ob* mice. Dimerizer treatment was initiated at 8 weeks of age and continued for 10 weeks. **(b)** Ventral aspect of representative FAT-ATTAC *ob/ob* (left) and *ob/ob* (right) mice, with exposed peritoneal cavity, showing absence of visible adipose in FAT-ATTAC mice treated for 6 weeks with dimerizer. **(c)** Increased daily food intake in FAT-ATTAC *ob/ob* mice with prolonged absence of adipose tissue (weeks 3–6 versus week 1, $P < 0.05$), and in comparison to *ob/ob* control mice. In both fed and fasted states, FAT-ATTAC *ob/ob* animals show increased body temperature **(d)**, plasma glucose **(e)** and plasma triglycerides **(f)**, compared to *ob/ob* littermates at 6 weeks of dimerizer treatment. **(g)** No difference in plasma nonesterified fatty acid (NEFA) concentration in FAT-ATTAC *ob/ob* and *ob/ob* animals. $*P < 0.05$. **(h)** Histology of *ob/ob* versus fatless FAT-ATTAC *ob/ob* livers.

4g), consistent with the adipocyte-specific expression of mouse $\beta 3$ adrenergic receptors¹¹. Upon cessation of dimerizer treatment, FAT-ATTAC mice showed a partial recovery of $\beta 3$ adrenergic-stimulated insulin secretion, indicating reconstitution of adipose tissue function (Fig. 4h). Similar to decreased serum adiponectin levels after adipose tissue reconstitution, the incomplete recovery of $\beta 3$ -stimulated insulin secretion may be the result of either a quantitative or qualitative difference in adipose depots. Future studies are necessary to determine whether an

increased dimerizer-free interval would lead to full recovery of adipocyte function. Further, although the mechanism mediating the effect of adipocyte $\beta 3$ stimulation on insulin secretion is unknown, and the possibility of intrinsic beta-cell dysfunction unaccompanied by obvious disruption of islet architecture or response to arginine challenge cannot be formally excluded, the effect lends additional support to a role for adipocytes in the regulation of insulin secretion and beta-cell function.

Relative contributions of adipose to the inflammatory state

Recent work from our group and others has shown that adipose tissue can mediate a considerable proportion of the systemic response to various inflammatory stimuli^{12–14}. To determine the contribution of adipocytes to circulating inflammatory cytokines and acute-phase reactants, we challenged fatless FAT-ATTAC mice and wild-type littermates with a sublethal dose (250 ng/g body weight) of lipopolysaccharide (LPS). In fatless mice, there is a significant reduction in the magnitude of the acute-phase response, as shown by a relative reduction in circulating serum amyloid A3 (Fig. 5a) and IL-6 (Fig. 5b) after LPS challenge. Furthermore, despite increased macrophage staining in white adipose depots in fatless FAT-ATTAC mice, we detected decreased circulating IL-6 in the basal (nonstimulated) state in the FAT-ATTAC mouse (15 pg/ml versus 55 pg/ml; Fig. 5c). The decreased basal inflammatory tone and reduced response to endotoxin may therefore reflect the loss of the adipocyte as a source of inflammatory cytokines. Alternatively, this may represent the loss of local adipocyte factors that stimulate adipose-resident macrophages, which are abundant in FAT-ATTAC mice

Diminished glucose-stimulated insulin secretion

Decreased serum insulin levels (75% and 66% decrease in fasting and fed insulin levels, respectively; Fig. 4a) persisted in the face of elevated plasma glucose in FAT-ATTAC *ob/ob* mice throughout the course of dimerizer treatment, despite increased insulin secretion in *ob/ob* littermates, which parallels progressive metabolic dysfunction in this mouse model (Fig. 4b). But 3 months after cessation of dimerizer treatment, serum insulin levels in FAT-ATTAC *ob/ob* mice returned to the severe hyperinsulinemic levels observed in control mice. An arginine challenge triggered similar levels of insulin secretion, and thus, a normal response to a glucose-independent challenge during the fatless state (Fig. 4c). To test whether the decrease in serum insulin resulted from a defect in glucose-stimulated insulin secretion (GSIS), we compared the effect of a glucose gavage on serum insulin levels in euglycemic FAT-ATTAC *ob/+* mice. Although the mice showed the expected increase in insulin levels in response to the glucose challenge before initiation of fat loss, once FAT-ATTAC *ob/+* mice became fatless, they did not increase serum insulin in response to an acute elevation of plasma glucose (Fig. 4d). There was no difference in pancreatic islet area or morphology in fatless FAT-ATTAC *ob/+* mice compared to controls (Fig. 4e,f). Similar findings were evident in the FAT-ATTAC *ob/ob* mice (data not shown). Together, these data suggest that the diminished GSIS in the fatless FAT-ATTAC mouse may result in part from the absence of an adipocyte-specific factor necessary for maximal GSIS. Furthermore, $\beta 3$ adrenergic-stimulated insulin secretion did not occur in the fatless FAT-ATTAC mouse (Fig.

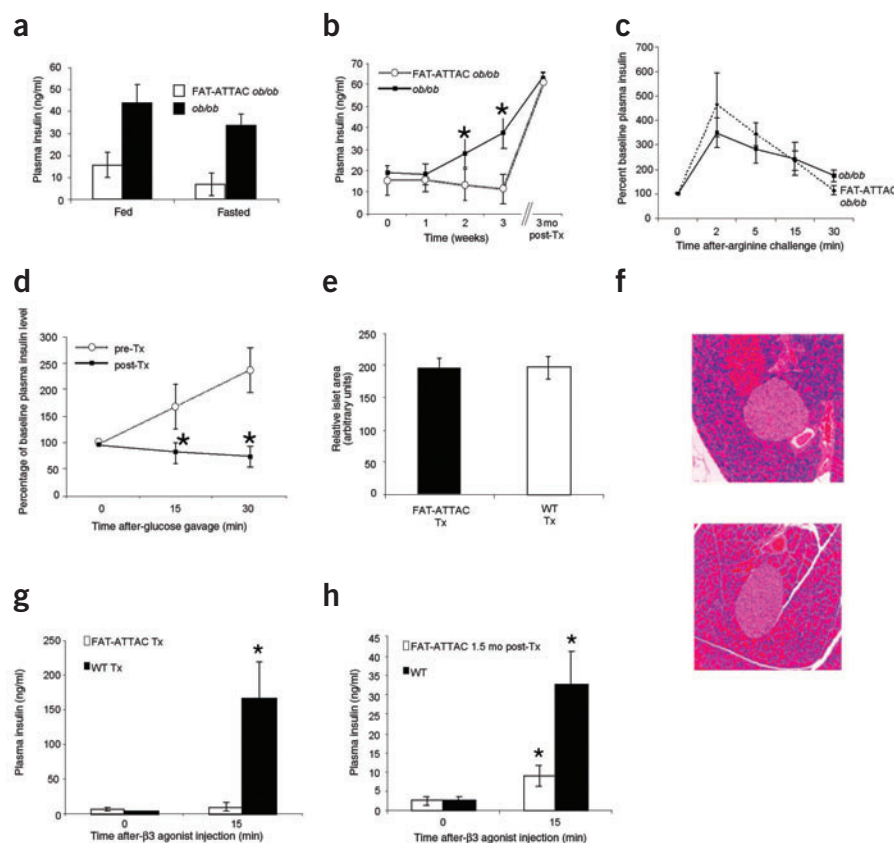


Figure 4 Defects in insulin secretion in FAT-ATTAC mice. **(a)** Paradoxically decreased plasma insulin levels in FAT-ATTAC *ob/ob* mice versus *ob/ob* mice in both fed and fasted states. **(b)** Increased plasma insulin in *ob/ob* mice from 2 weeks after initiation of dimerizer treatment (~10 weeks of age), with no increase in insulin secretion in FAT-ATTAC *ob/ob* animals. **(c)** Similar kinetics and relative magnitude of insulin secretion after *in vivo* administration of 1 mg/g body weight arginine in both FAT-ATTAC *ob/ob* and *ob/ob* animals. **(d)** After a glucose gavage, FAT-ATTAC mice on dimerizer treatment for 4 weeks do not show GSIS as assessed through insulin excursion compared to these same mice before dimerizer treatment. **(e)** Quantification of islet size shows that the average islet area in FAT-ATTAC mice is similar to wild-type mice after 4 weeks of dimerizer treatment. **(f)** Histological sections of pancreatic islets further show the similarities in islet size between wild-type (top) and FAT-ATTAC (bottom) mice after 4 weeks of dimerizer treatment. **(g)** Increases in serum insulin levels 15 min after β 3 agonist injection are markedly higher in wild-type mice, but FAT-ATTAC mice treated with dimerizer for 4 weeks are unresponsive. **(h)** One and a half months after cessation of dimerizer treatment, FAT-ATTAC mice show responsiveness to β 3 agonist injection by increasing insulin secretion. But the magnitude of the response in FAT-ATTAC mice is less than that of wild-type controls. **P* < 0.05. WT, wild-type; Tx, treatment; mo, month.

(Supplementary Fig. 1 online), to produce inflammatory cytokines in response to acute or chronic inflammatory challenges¹⁵.

DISCUSSION

The ability to induce widespread and reversible loss of adipocytes at any time during development represents a major technological advance. The initial characterization of the FAT-ATTAC mouse presented here shows that this mouse is unequivocally fatless. We present data detailing the necessity of adipose tissue for maximal GSIS, evidence for requirements of adipocyte-derived factors other than leptin in the maintenance of food intake and energy expenditure, as well as the effects of the adipocyte on the systemic inflammatory state. These findings were obtained in a model that does not possess the added secondary complications of long-term insulin resistance present in other lipotrophic mouse models. The reversibility of our lipotrophic mouse model provides a unique tool to study *de novo* adipocyte differentiation *in vivo*, as well as the *in vivo* characterization of many inhibitors initially identified in 3T3-L1 adipocyte differentiation in cell culture. The FAT-ATTAC mouse model may also be useful to address the contribution of local adipocyte-derived factors to morbidity and mortality in pathologic states, and will surely highlight the importance of the adipocyte in the regulation of multiple physiological functions.

METHODS

Materials and animals. C57BL/6J *ob/+* animals were supplied by Jackson Laboratories. We maintained mice on a 12-h light-dark cycle and standard chow diet. We administered AP21087 (Ariad Pharmaceuticals) or vehicle (4% ethanol, 10% PEG-400, 1.75% Tween-20 in water) every 3 d by intraperitoneal injection at a dose of 0.2 μ g/g body weight unless otherwise indicated. Animals were between 6 and 8 weeks of age upon the initiation of dimerizer treatment.

All animal experimental protocols were approved by the Institute for Animal Studies of the Albert Einstein College of Medicine.

Cell culture. Subconfluent 3T3-L1 mouse fibroblasts were cotransfected with the FAT-ATTAC transgenic targeting vector and pCB7 to allow selection for hygromycin resistance. We manually selected hygromycin-resistant stable clones, propagated them and differentiated them as previously described¹⁶. Cell lines expressing caspase 8–FKBP–FLAG were identified by immunoblotting using M2 FLAG-specific antibody (Sigma-Aldrich); these cells were allowed to fully differentiate into adipocytes, then treated for 2 d with either vehicle (0.5% ethanol) or AP20187 (100 nM).

Generation of FAT-ATTAC transgenic mice. Activated human caspase 8 (p20-p10 transgene corresponding to Ser217–Asp479) was amplified, 3' FLAG-tagged and subcloned in-frame as a *SpeI*–*BamHI* fragment into vector pC4M–Fv2E (Ariad Pharmaceuticals) to generate a cDNA encoding a myristoylated caspase 8–FKBP–FKBP fusion protein. A further PCR amplification step of the entire cDNA including downstream rabbit β -globin intron, polyadenylation sequence and 3' untranslated region for transgene stability with flanking *NotI* sites allowed subcloning of the entire 2.8 kb sequence into a pBluescriptKS+–based vector containing the minimal 5.4 kb of the *Fabp4* promoter required for adipocyte-specific expression (provided by B. Spiegelman, Dana Farber Cancer Institute). pBluescriptKS+ sequence was excised with a *SacII*/*SalI* digest and the remaining 8.2 kb linearized plasmid injected into pronuclei of fertilized mouse eggs derived from FVB animals. Transgenic progeny were identified by Southern analysis using an 800 bp fragment of human caspase 8 as a probe; subsequent generations were screened by PCR of tail-derived genomic DNA using human caspase 8–specific primers (forward: 5'-GAAAGTGCCCAAACTTCACAGCATTAGG-3'; reverse: 5'-CTTGTCATCCTTGTTGCTTACTTCATAG-3'), giving rise to a 600 bp product. Transgene expression was verified by northern blot and RT-PCR analysis of total RNA isolated from white adipose and other tissues using Trizol (Life Technologies). Western blot analysis of 20 μ g of total protein from tissue lysates in PBS + 2 mM phenylmethylsulfonylfluoride and protease-inhibitor cocktail (Boehringer-Mannheim) was performed as described previously¹⁷.

Figure 5 Functional adipose is required for normal basal and LPS-stimulated inflammatory tone. (a) Decreased LPS-stimulated serum amyloid A3 in lipotrophic FAT-ATTAC *ob/ob* mice relative to control *ob/ob* littermates or FAT-ATTAC *ob/ob* animals treated with vehicle. Decreased LPS-stimulated (b) and basal (c) serum IL-6 in FAT-ATTAC *ob/ob* animals treated with dimerizer for 6 weeks. Basal levels of serum IL-6 in fatless mice are roughly 25–30% of vehicle-treated FAT-ATTAC *ob/ob* littermates.

Assays. We measured serum values for glucose using FastBlue B glucose assay (Sigma-Aldrich), triglycerides using Infinity Triglycerides kit (ThermoDMA), free fatty acids using NEFA-C kit (Wako Chemicals), insulin using rat-sensitive insulin RIA kit (LINCO Research), and adiponectin using mouse adiponectin RIA kit (LINCO Research). Other adipokines were measured using LINCOPLEX reagents. All metabolic parameters measured in the fed state were in animals that had food withdrawn at 4 PM, replaced at 7 PM and assayed at 11 PM; for fasted-state measurements, we assayed animals at 10 AM after a 16-h fast. We performed oral glucose-tolerance tests in animals that did not have access to food for 2 h before administration of 2.5 mg/g body weight glucose load by oral gavage, and during the course of the study. We performed arginine-tolerance tests by intraperitoneal administration of a single 1 mg arginine/g body weight dose in animals fasted for 16 h. The effect of $\beta 3$ agonists on insulin secretion was tested by intraperitoneal injection of CL316423 (1 μ g/g body weight) into fed animals. We performed LPS challenge by administration of 250 ng/g body weight of *E. coli*-derived LPS (Sigma-Aldrich) by intraperitoneal injection; we analyzed blood obtained through tail vein for serum IL-6 by Quantikine IL-6 ELISA (R&D Systems) and serum amyloid A3 (ref. 12) by quantitative western blotting using an 125 I-derivatized secondary goat rabbit-specific antibody (Amersham) and analyzed it with a Phosphorimager (Molecular Dynamics). For thiazolidinedione efficacy studies, we administered daily doses to FAT-ATTAC *ob/ob* animals and *ob/ob* littermates for 10 d by oral gavage of 10 mg/kg/d rosiglitazone in 0.25% carboxymethylcellulose.

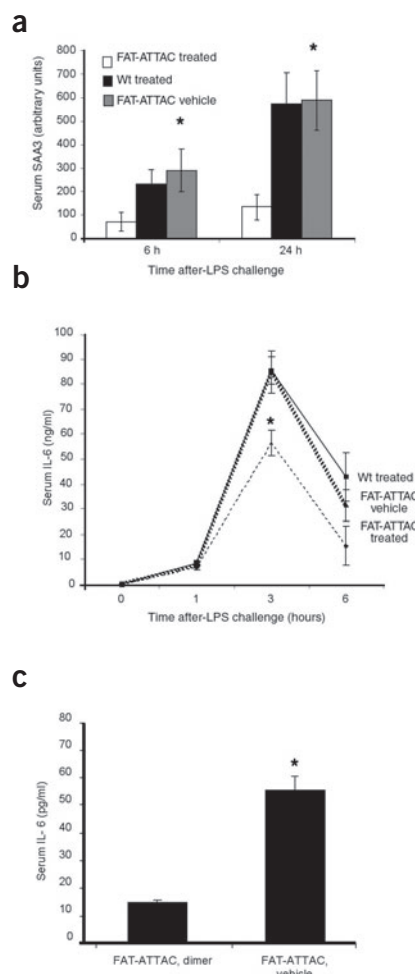
Immunohistochemistry. Freshly isolated tissues were fixed with phosphate-buffered formalin and then embedded in paraffin wax. We incubated sections (5 μ m) with affinity-purified rabbit antibody to perilipin (Affinity BioReagents) or rat antibody to F4/80. After washing in PBS, we incubated slides with biotinylated goat antibody to rabbit IgG or goat antibody to rat IgG at 5 μ g/ml (Vector Laboratories). Slides were developed using a peroxidase detection kit (Vector Laboratories) and counterstained with hematoxylin (Sigma-Aldrich). We performed TUNEL staining and cleaved caspase 3 immunohistochemical detection on paraffin-embedded sections as previously described¹⁸. We performed oil red O staining on 10 μ m sections of liver frozen in optimal tissue cutting support medium (Tissue-Tek). Hematoxylin was used as a counterstain. Pancreatic islet area was quantified using US National Institutes of Health Image J Software. We sampled at least 80 islets per group.

Magnetic resonance imaging. All images were obtained using a 9.4 Tesla imaging system. We anesthetized mice with isoflurane (1.5% in O₂) delivered using a nose cone. To quantitatively assess whole body fat and water, we subjected each mouse to a four-scan pulse-acquire sequence in a 40 mm 1 H coil, and obtained and analyzed spectra, including the water and fat peaks. For imaging, several datasets of eight slices of 2 mm thickness spanning the whole body were acquired using a routine spin echo pulse sequence (18 ms echo time, 400 ms repetition time and two-signal averages).

Note: Supplementary information is available on the Nature Medicine website.

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COMPETING INTERESTS STATEMENT

The authors declare that they have no competing financial interests.

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